Assignment 3 :
Using FVM with SIMPLE algorithm to simulate steady Lid Driven Cavity

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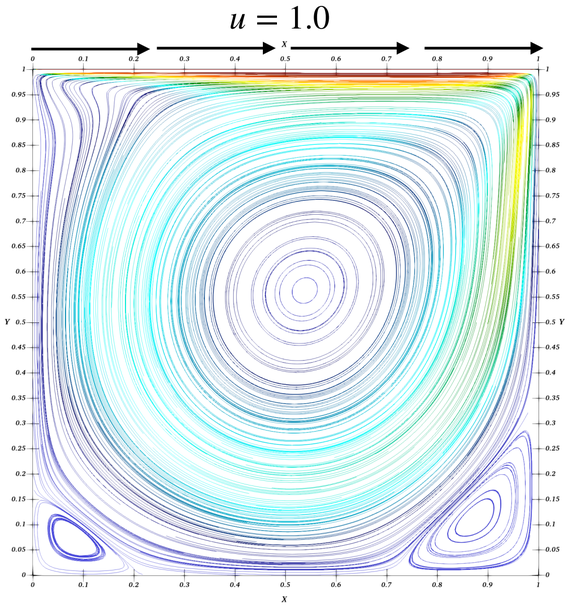
1. Problem descriptions

Governing Equation : Navier-Stokes Equations

$$\begin{cases} \vec{\nabla} \cdot \vec{v} = 0 \\ \vec{\nabla} \cdot \rho\vec{u} \otimes \vec{u} = -\vec{\nabla} p + \mu \nabla^2 \vec{u} \end{cases}$$



▲ Fig. 1.1 Lid Driven Cavity at Re = 1000

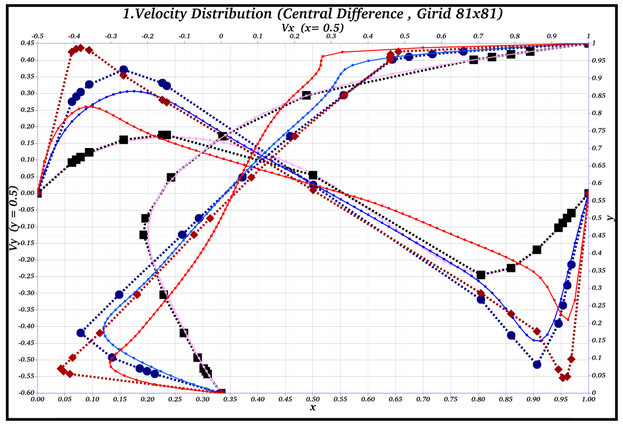
(1)	Top : u = 1 , v = 0
(2)	Bottom : u = 0 , v = 0
(3)	Left : u = 0 , v = 0
(4)	Right : u = 0 , v = 0

▲ Fig. 1.2 Boundary Condition

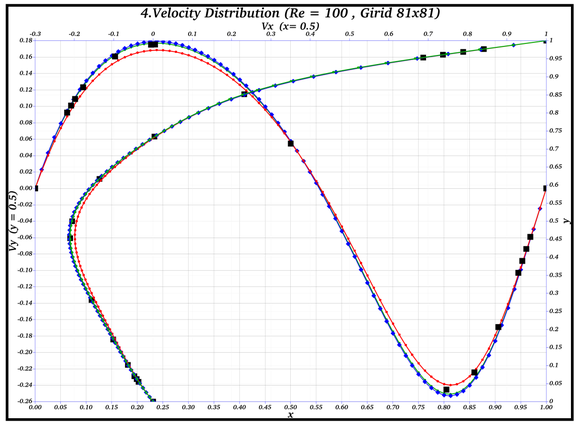
(1) Compute velocity distribution for x part and y part and pressure field

(2) Centerline velocity comparison under the same discretization scheme

(3) Centerline velocity comparison under the same R_e number



▲



▲

Fig. 1.3 Central Difference scheme

Fig. 1.4 Velocity distribution Re = 100

2. Discretize the governing equation and using staggered grid

1. Integration form of the 2D momentum equation and continuity equation

$$\frac{\partial \rho u}{\partial x} + \frac{\partial \rho u}{\partial x} = 0$$

$$\frac{\partial \rho u \cdot u}{\partial x} + \frac{\partial \rho v \cdot u}{\partial y} = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial x^2}$$

Gauss-Divergence Theorem

$$\oint_{\partial\Omega} d\vec{a} \cdot \vec{u} = 0$$

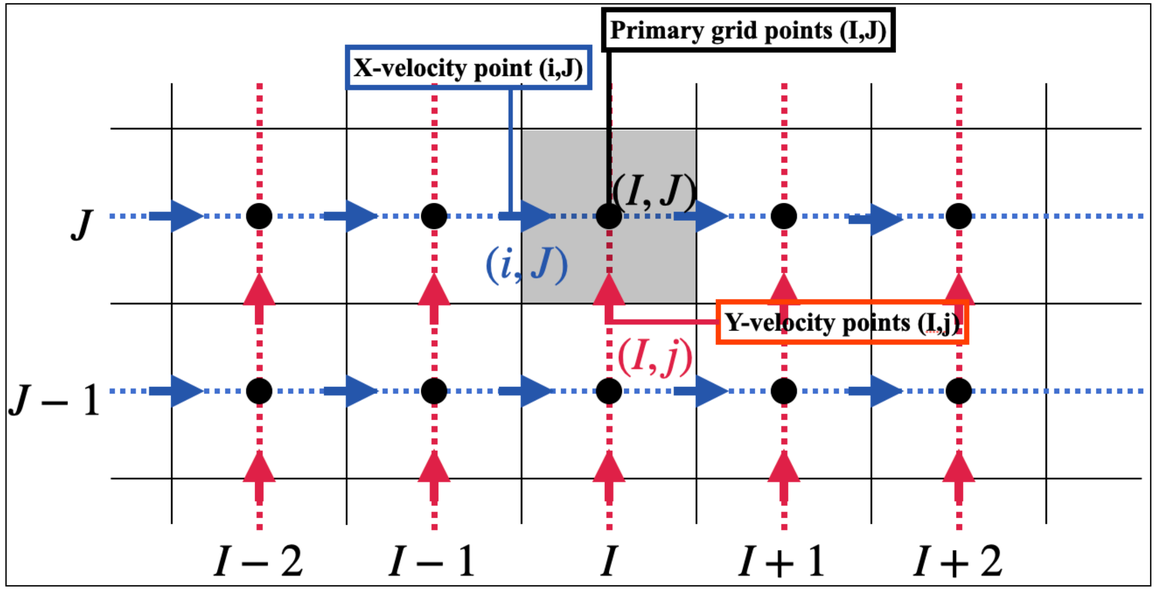
$$\oint_{\partial\Omega} dy(u^2 + uv) = \oint_{\partial\Omega} dy \frac{(-p)}{\rho} + \frac{1}{R_e} \oint_{\partial\Omega} dy \frac{\partial u}{\partial x} + dx \frac{\partial u}{\partial y}$$

2. The staggered grid

I define three types of grid points to calculate three variables in the governing equations:

1. Primary points: scalar variables (e.g., p) , (I, J)
2. x-velocity points: u , (i, J)
3. y-velocity points: v , (I, j)

This configuration is referred to as a backward staggered grid.



▲ Fig. 2.1 Let discuss why using the staggered grid

2. Discretize the governing equation and using staggered grid

2.staggered grid : why ? Non-physical

When we discretize the equations using central differences, certain problems arise regardless of whether FVM or FDM is employed.

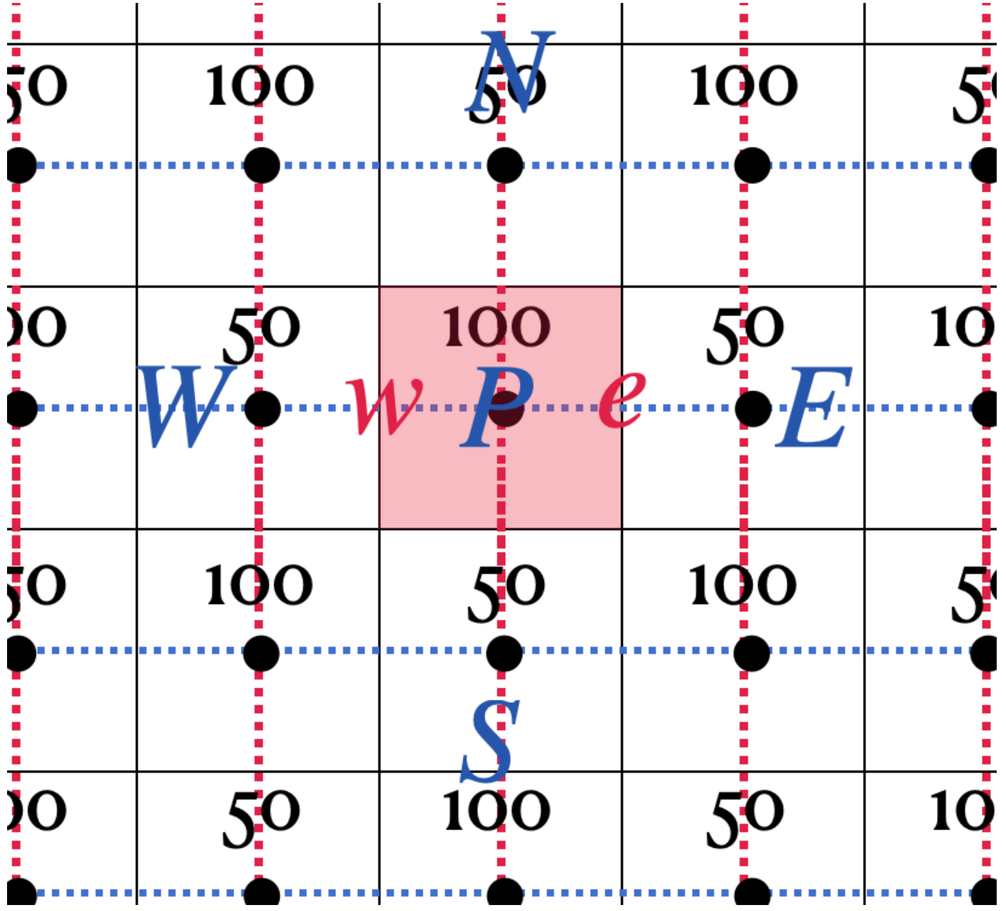
To understand the reason , let revisit the discretization process :

$$-\frac{1}{\rho}\oint_{\partial\Omega} dy \cdot p \approx -\frac{1}{\rho}(p_e - p_w) \approx -\frac{1}{2\rho}(p_E - p_W) \qquad (2.1)$$

Central Difference scheme

As the result , this pressure field would give the same (zero) momentum source in the discretized equations as a uniform pressure form.

How can we solve it ?



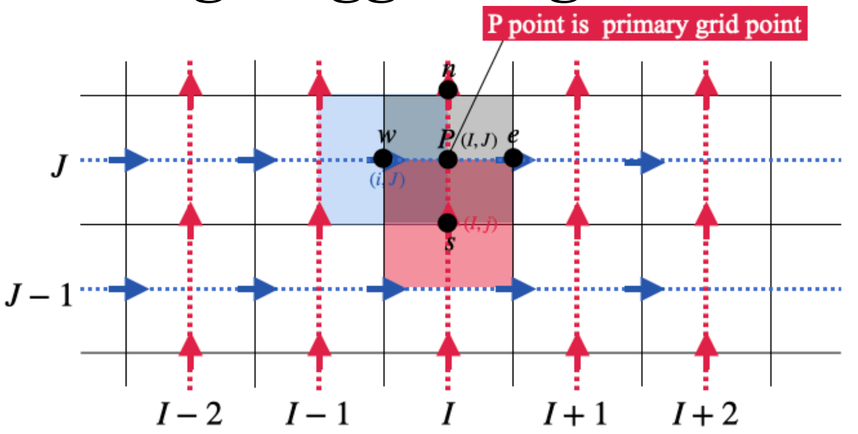
▲ Fig. 2.2 Let discuss why using the staggered grid

2. Discretize the governing equation and using staggered grid

3.Discretizing the continuity equation

(1)center of the controlled volume : primary point P

(2)Position : (I, J)



coefficient of convection	$P = (I, J)$
F_e^p	$A_e u_{i+1,J}$
F_w^p	$A_w u_{i,J}$
F_n^p	$A_n v_{I,j+1}$
F_s^p	$A_s v_{I,j}$

▲ Fig. 2.3 discretizing the continuity equation

3.1 integration form of continuity equation :

$$\oint_{\partial\Omega} d\vec{a} \cdot \vec{u} = 0 \qquad (2.2)$$

3.2 discrete continuity equation :

$$F_e^p - F_w^p + F_n^p - F_s^p = 0 \qquad (2.3)$$

Where

(1) $F_e^P = A_e u_E = A_e u_{i+1,J}$

(2) $F_w^P = A_w u_P = A_e u_{i,J}$

(3) $F_n^P = A_n v_N = A_n v_{I,j+1}$

(4) $F_s^P = A_s v_P = A_s v_{I,j}$

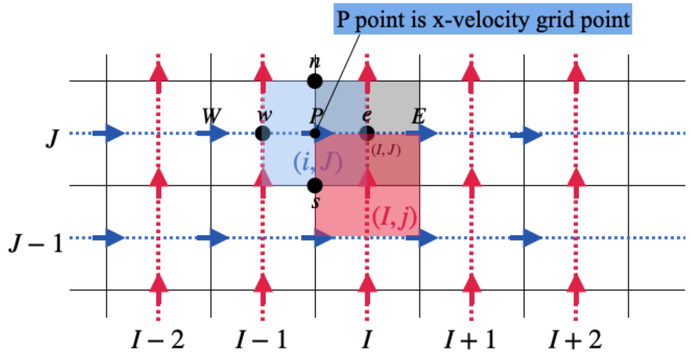
▲ Fig. 2.4 coefficients of discrete continuity equation

2. Discretize the governing equation and using staggered grid

4.Discretizing the x-momentum equation

(1)center of the controlled volume : x-Velocity point

(2)Position : (i, J)



coefficient of convection	$P = (i, J)$
F_e^u	$A_e \frac{u_E + u_P}{2}$
F_w^u	$A_w \frac{u_P + u_W}{2}$
F_n^u	$A_n \frac{v_{I,j+1} + v_{I-1,j+1}}{2}$
F_s^u	$A_n \frac{v_{I,j} + v_{I-1,j}}{2}$

▲ Fig. 2.5 discretizing the x-momentum equation

4.1 integration form of x-momentum equation :

$$\oint_{\partial\Omega} dyu \cdot u + dxv \cdot u = \int_{\Omega} dr^3 \left(-\frac{1}{\rho}\frac{\partial p}{\partial x} \right) + \frac{1}{R_e} \oint_{\partial\Omega} dy \frac{\partial u}{\partial x} + dx \frac{\partial u}{\partial y} \tag{2.4}$$

4.2 General discrete of the integral x-momentum equation :

$$F_e^u u_e - F_w^u u_w + F_n^u u_n - F_s^s u_s = \frac{1}{\rho} \frac{-\left(p_{I,J} - p_{I-1,J}\right)}{\delta x} \delta x \delta y$$

$$-u_P \left(D_w + D_e + D_s + D_n\right) + u_W \left(D_w\right) + u_E \left(D_e\right) + u_S \left(D_s\right) + u_N \left(D_n\right) \tag{2.5}$$

3. SIMPLE algorithm

1. Problem

Discrete governing equation : How can we get the correct pressure value ?

Continuity Equation : $F_e^p - F_w^p + F_n^p - F_s^p = 0$ (3.1)

X-momentum Equation : $a_P u_P + a_{EE}u_{EE} + a_E u_E + a_W u_W + a_{WW}u_{WW} + a_{NN}u_{NN} + a_N u_N + a_S u_S + a_{SS}u_{SS}$

$$= \frac{1}{\rho}\frac{-(p_P - p_w)}{\delta x}\delta x \delta y \quad (3.2)$$

2. Method: Predictor-Corrector Approach

Correct value = Guessed value + Correction value $p = p' + p^\star$

(1) Advantage : The pressure correction equation can be used to update the pressure correction field independently, and this equation is equivalent to the continuity equation.

(2) Disadvantage : The number of variables increases from 3 to 9.

$$[u', v', p'] \longrightarrow [u, v, p, u', v', p', u^*, v^*, p^*]$$

3. SIMPLE algorithm

3. Equations : Three main equations in predict-correction method

(1) Pressure Correction Equation (Pressure Poisson Equation) :

$$\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i+1,J}}+\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i,J}}+\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j+1}}+\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j}}\right)p'^{\,N+1}_{I,J}$$

$$=\left(\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j}}\right)p'^{\,N}_{I,J-1}+\left(\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j+1}}\right)p'^{\,N}_{I,J+1}+\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i,J}}\right)p'^{\,N}_{I-1,J}+\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i+1,J}}\right)p'^{\,N}_{I+1,J}$$

$$+u^{*\,N+\frac{1}{2}}_{i,J}-u^{*\,N+\frac{1}{2}}_{i+1,J}+v^{*\,N+\frac{1}{2}}_{I,j}-v^{*\,N+\frac{1}{2}}_{I,j+1}=0 \qquad (3.3)$$

Meaning : Given the predicted velocity, obtain the predicted pressure.

$[u^*, v^*]$ → Eq. (3.3) → $[p']$

3. SIMPLE algorithm

3. Equations : Three main equations in predict-correction method

(2) predicted momentum equation :

$$(a_P)_{i,J}^{*\,N} u_{i,J}^{*\,N+\frac{1}{2}} + (a_{WW})_{i,J}^{*\,N} u_{i-2,J}^{*\,N} + (a_W)_{i,J}^{*\,N} u_{i-1,J}^{*\,N} + (a_E)_{i+1,J}^{*\,N} u_{i+1,J}^{*\,N} + (a_{EE})_{i+2,J}^{*\,N} u_{i+2,J}^{*\,N}$$
$$+(a_{SS})_{i,J}^{*\,N} u_{i,J-2}^{*\,N} + (a_S)_{i,J}^{*\,N} u_{i,J-1}^{*\,N} + (a_N)_{i,J}^{*\,N} u_{i,J+1}^{*\,N} + (a_{NN})_{i,J}^{*\,N} u_{i,J+2}^{*\,N}$$
$$= \frac{-\left(p_{I,J}^{*\,N} - p_{I-1,J}^{*\,N}\right)}{\rho \delta x} \delta x \delta y \qquad (3.4)$$

Meaning : Given the predicted pressure , obtain the predicted velocity .

$[p^*]$ → Eq. (3.4) → $[u^*, v^*]$

3. SIMPLE algorithm

3. Equations : Three main equations in predict-correction method

(3) Correct momentum equation :

$$(a_P)^{*\,N+\frac{1}{2}}_{i,J}\, u'^{N+1}_{i,J} = \frac{-\left(p'^{N+1}_{I,J} - p'^{N+1}_{I-1,J}\right)}{\rho\delta x}\delta x\delta y \tag{3.5}$$

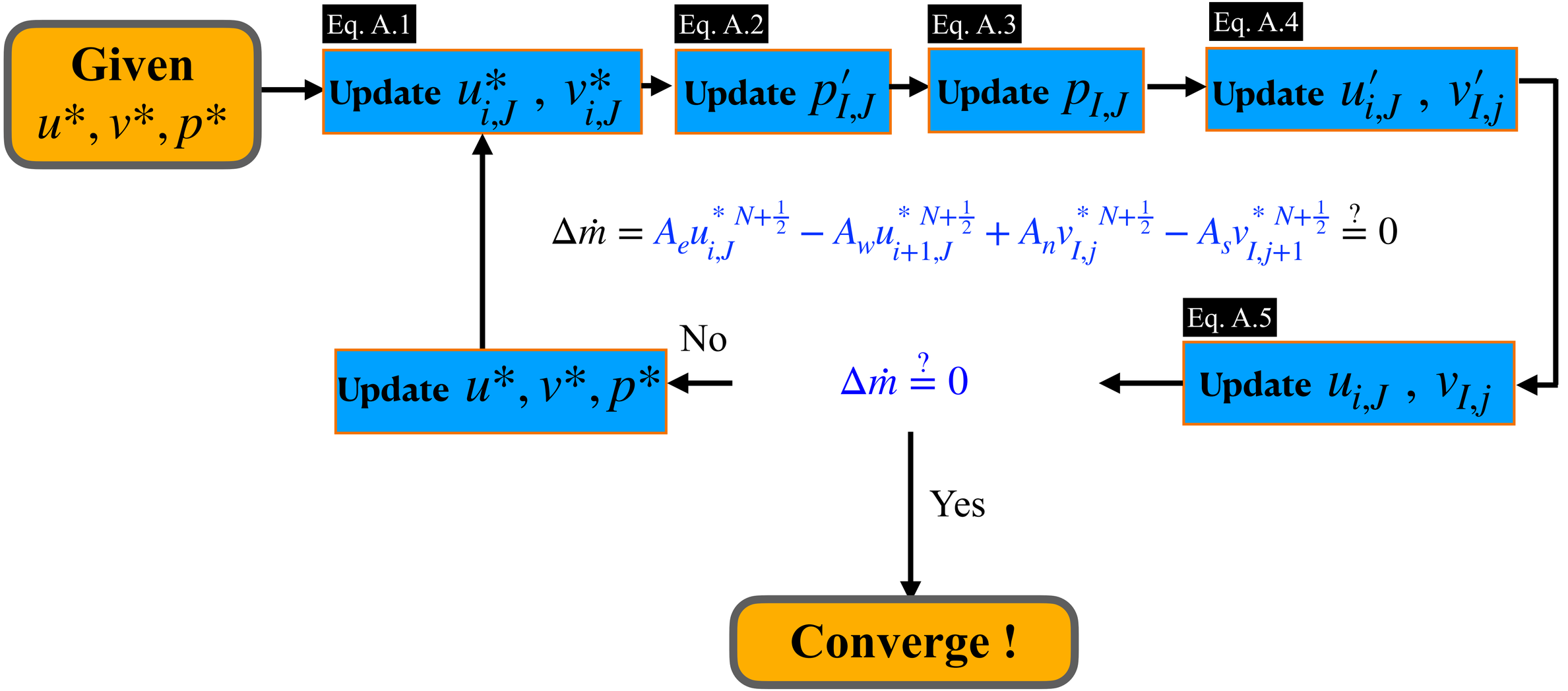
Meaning: Given the pressure correction field, calculate the velocity correction field.

$$[p'] \longrightarrow \boxed{\text{Eq. (3.5)}} \longrightarrow [u', v']$$

Point about equation (3.5) : Equation (3.5) would originally take a complex form similar to Equation (3.4). However, in the SIMPLE algorithm, Patankar introduced a crucial simplification by omitting the contributions from neighboring grid points.

3. SIMPLE algorithm

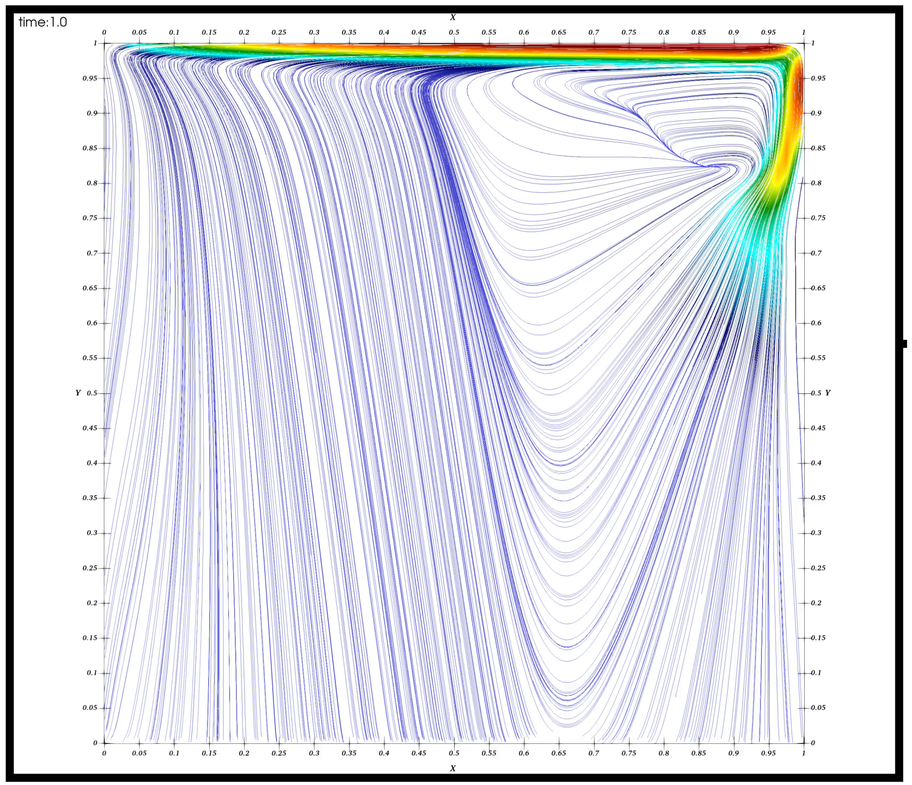
4. Algorithm

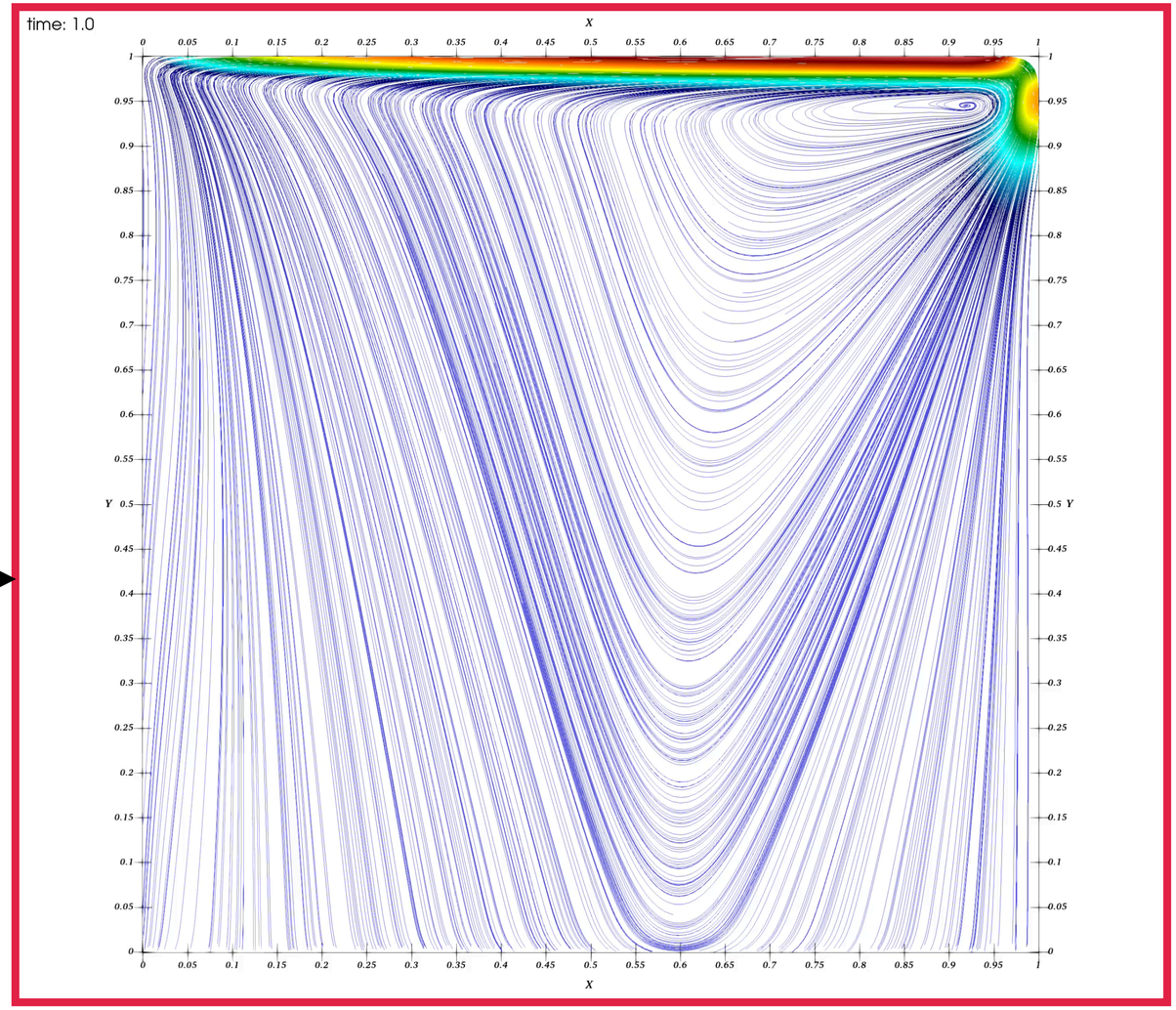


Appendix .1

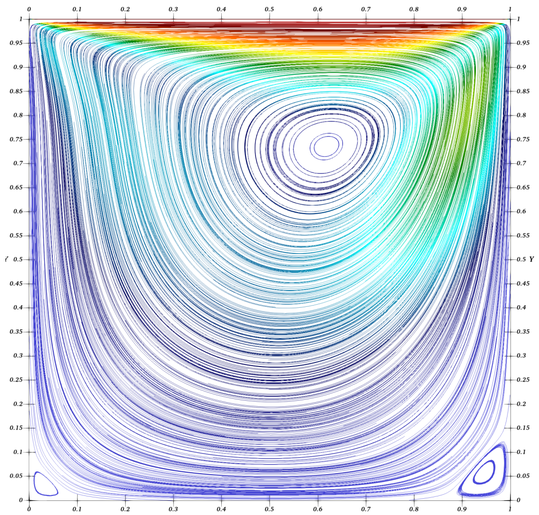
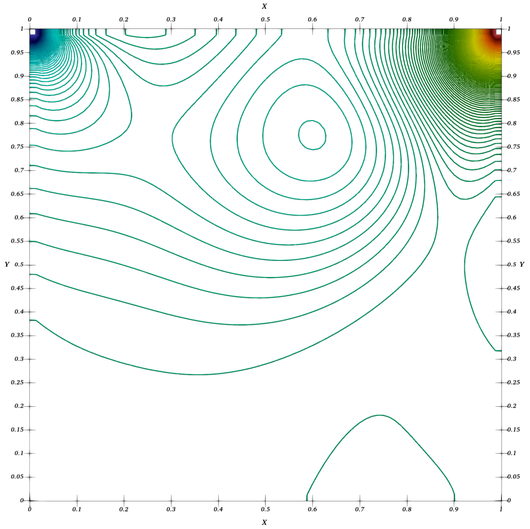
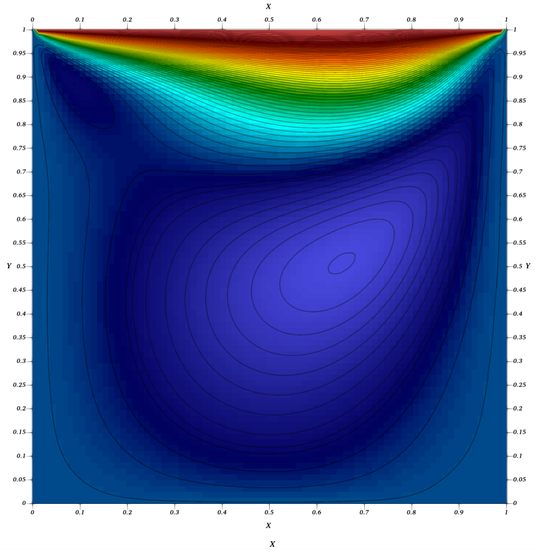
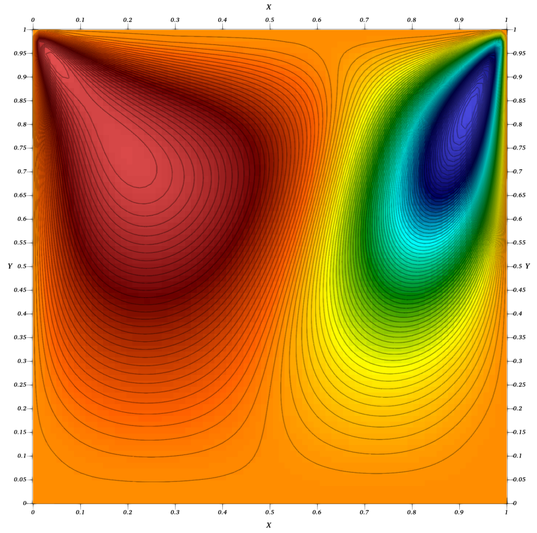
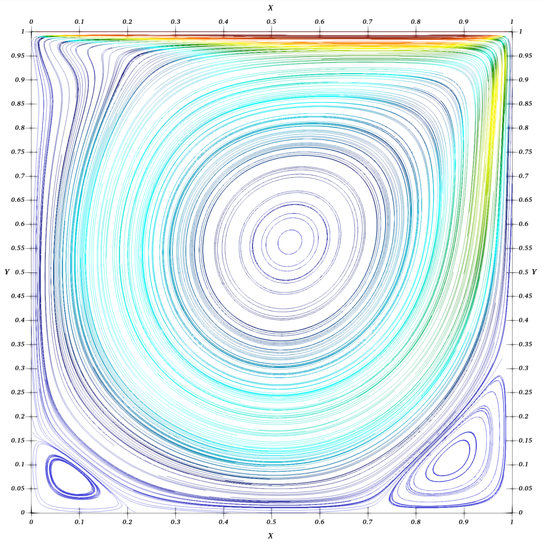
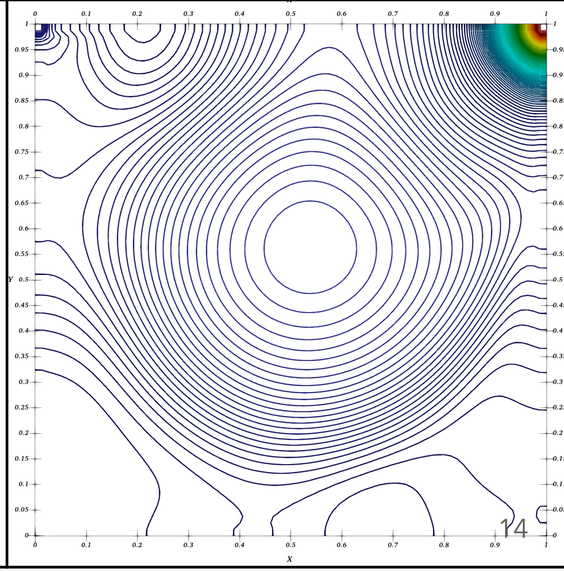
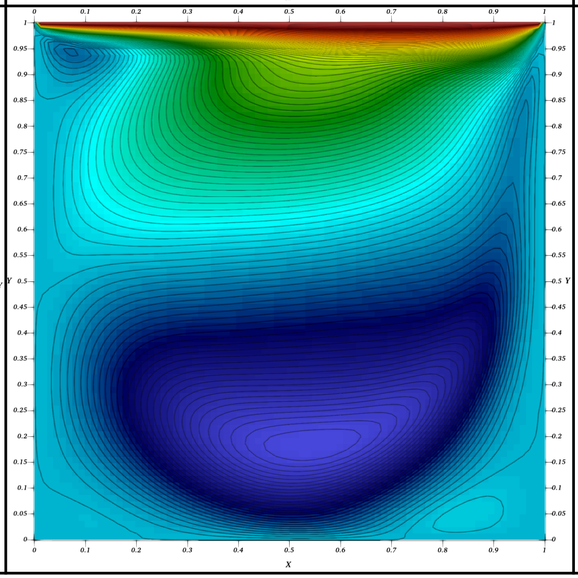
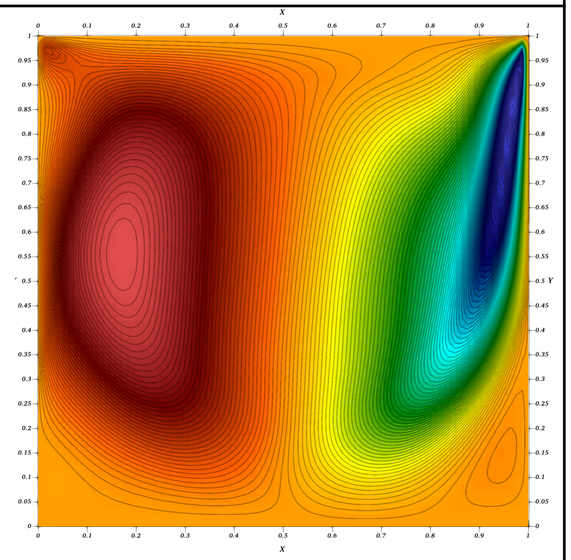
4. My Failure Case

1. $R_e = 5000$
2. Scheme : TVD-MUSCL scheme
3. Improvement Plan ：deferred correction

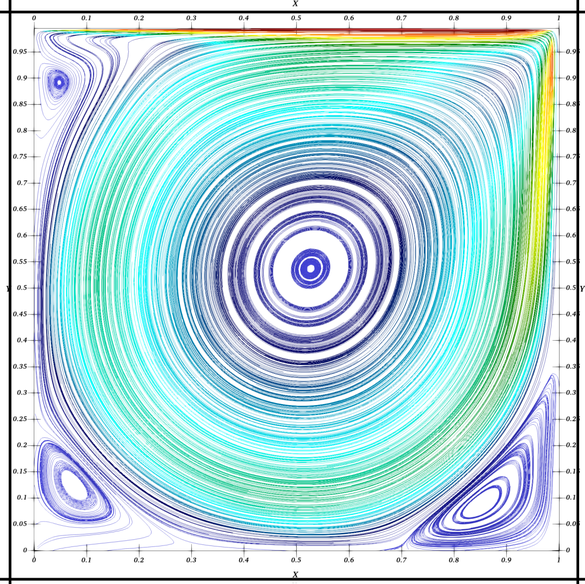
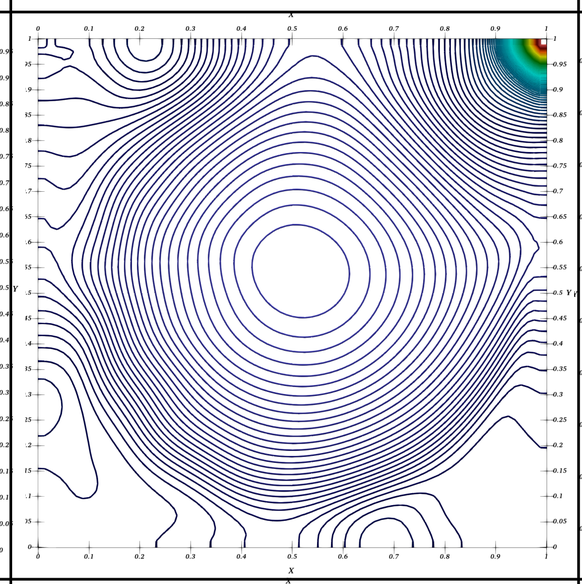
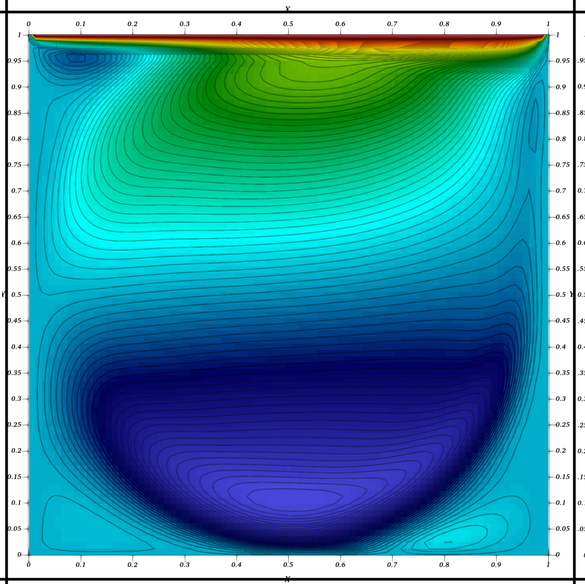
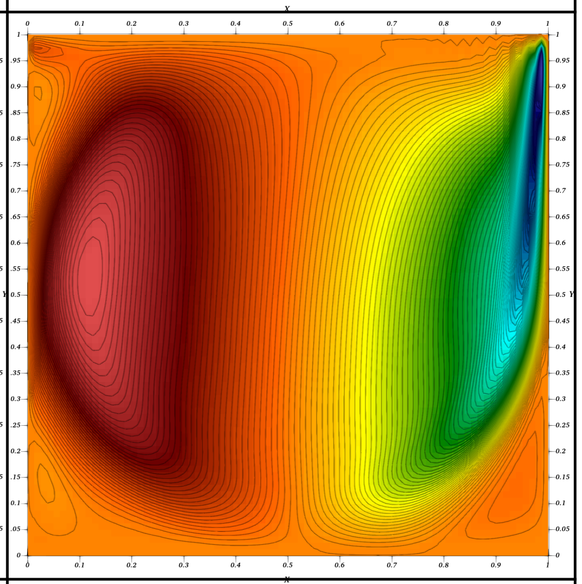
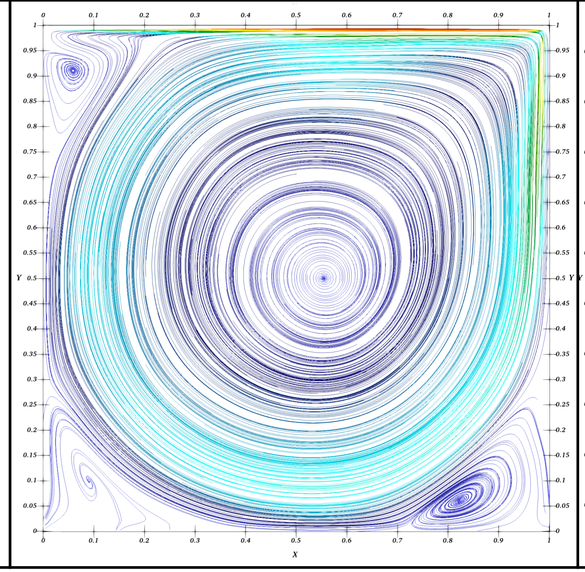
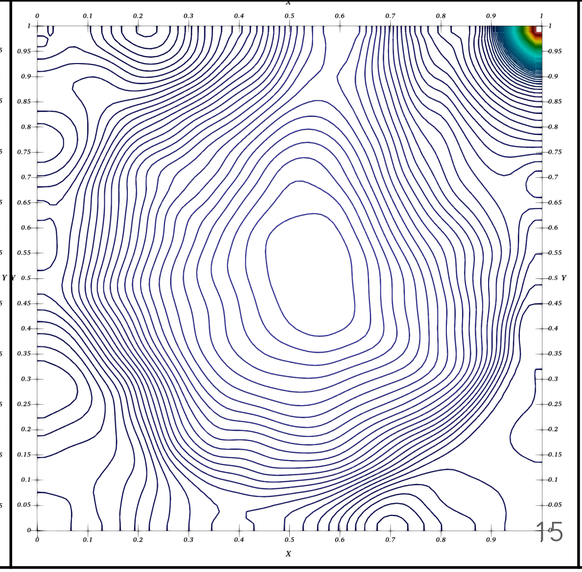
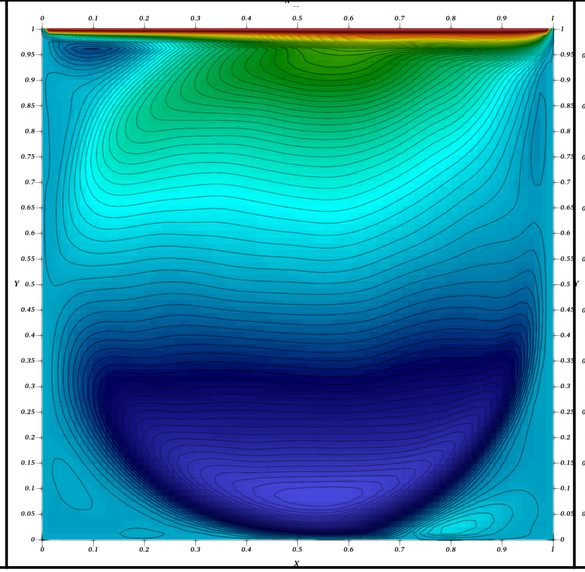
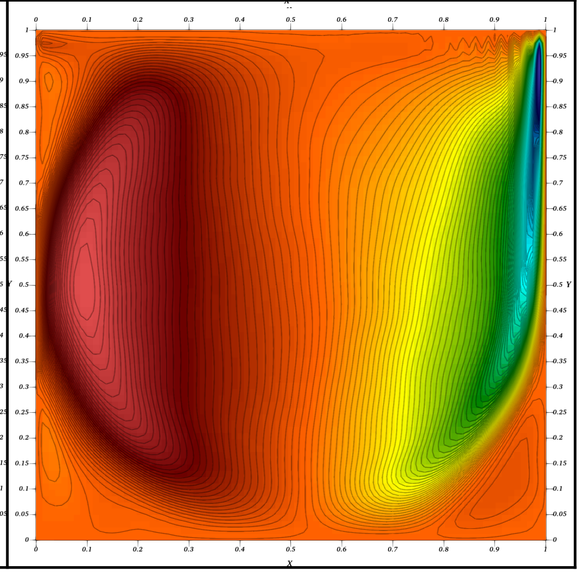




5. Result and Discussion

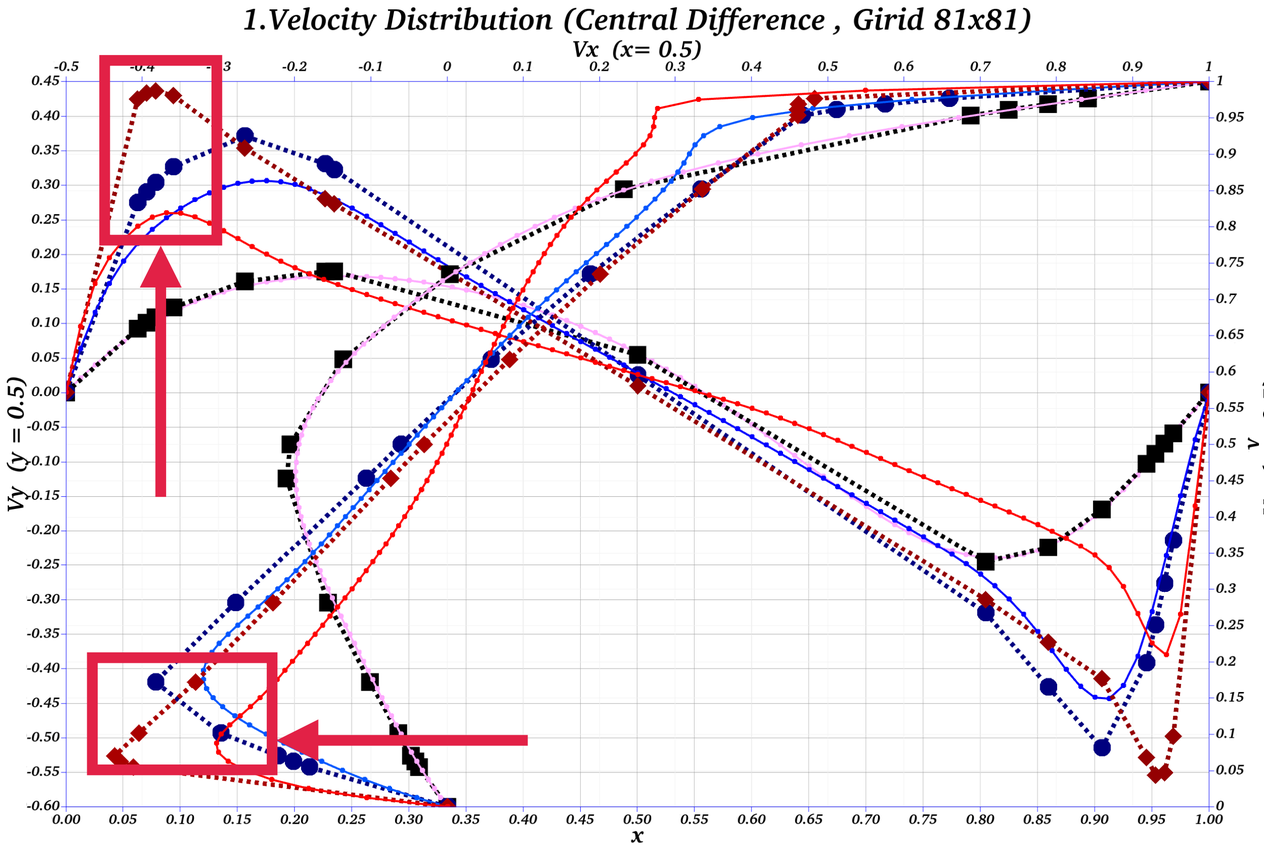
Re	streamline	Pressure field	X-velocity field	Y-velocity field
100				
1000				

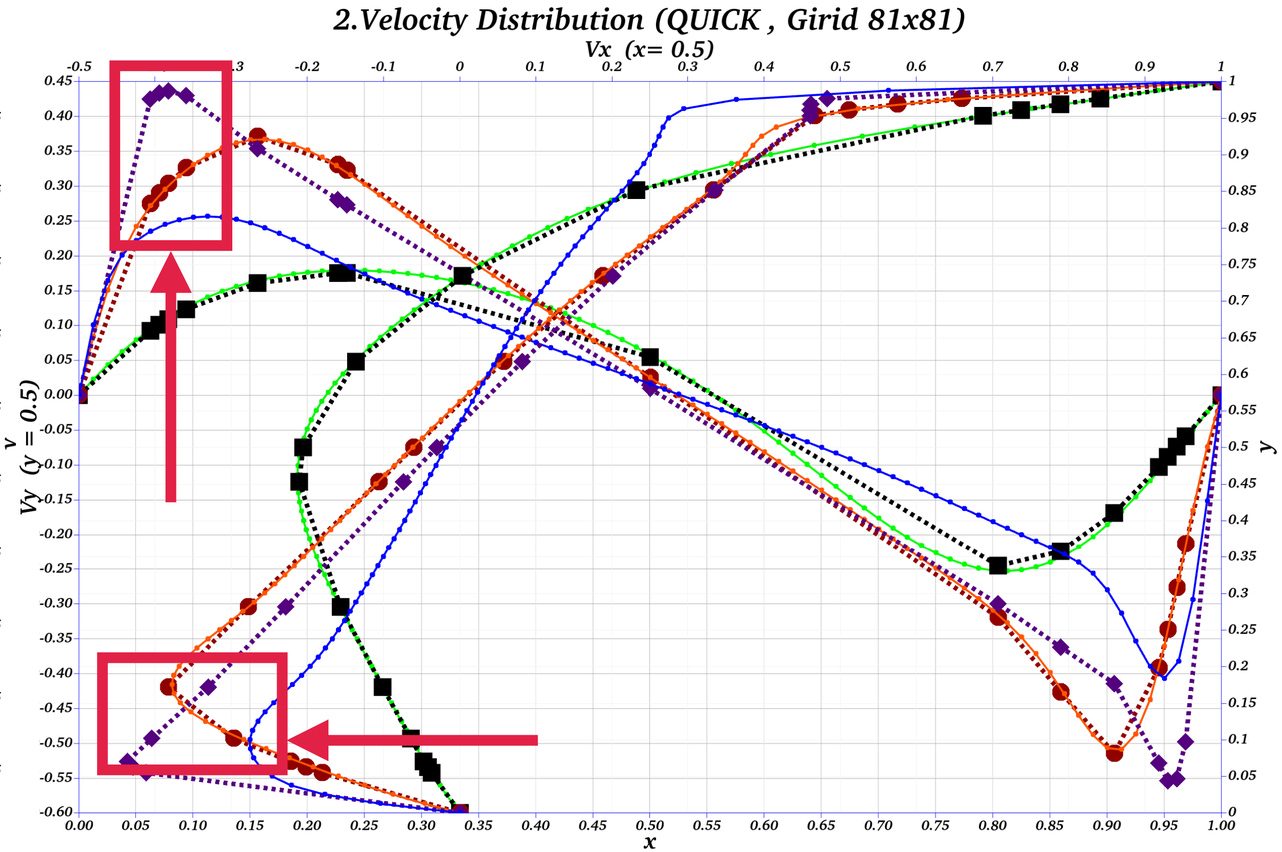
5. Result and Discussion

Re	streamline	Pressure field	X-velocity field	Y-velocity field
3000				
5000				

5. Result and Discussion

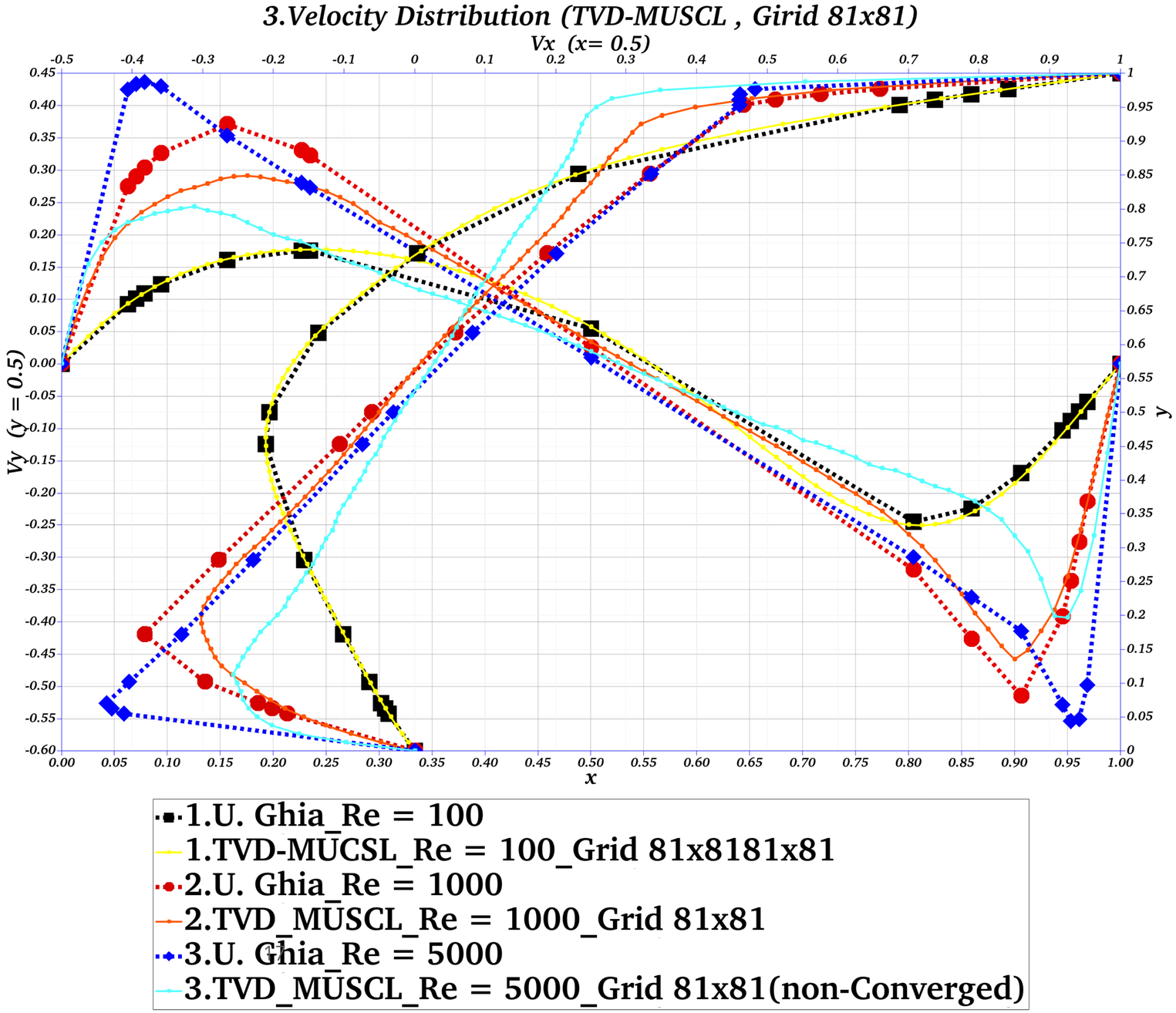
(1) Centerline velocity comparison under the same discretization scheme





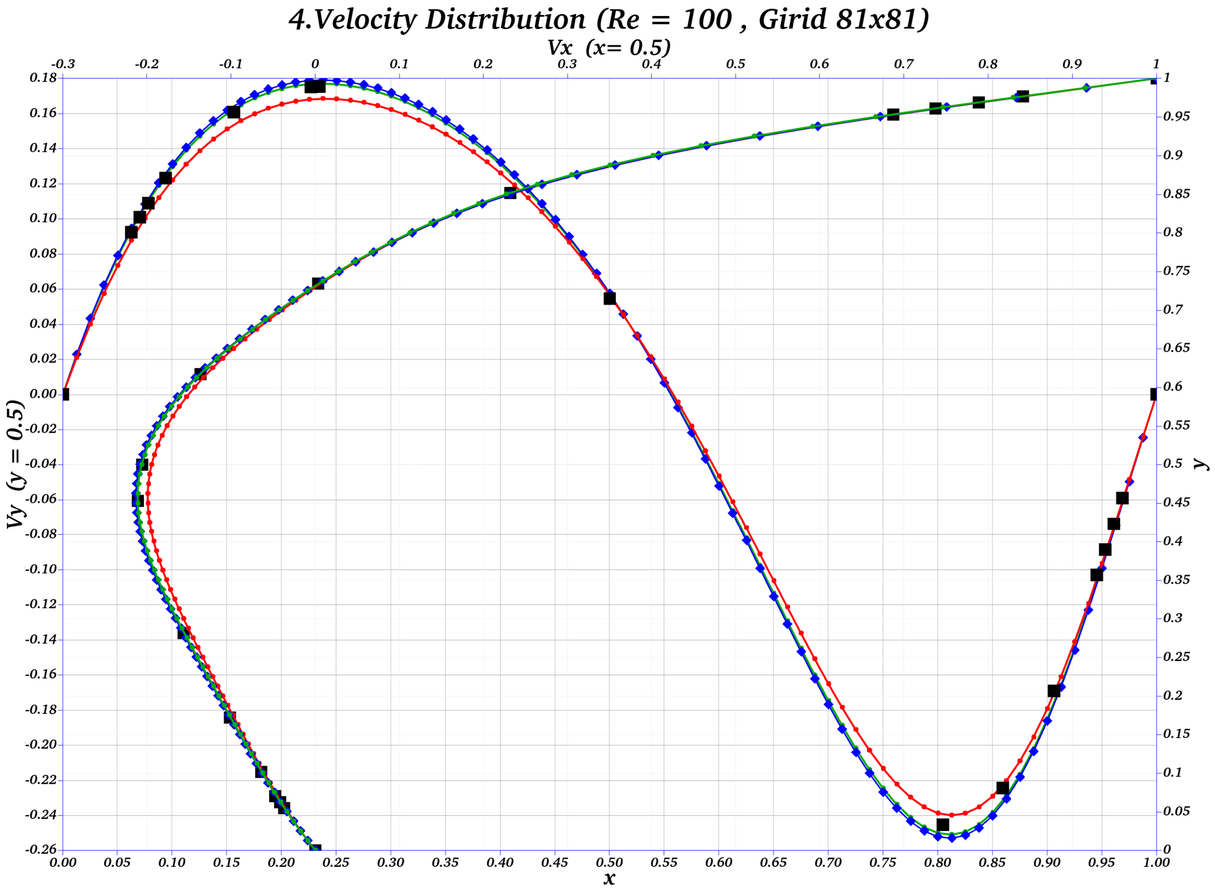
5. Result and Discussion

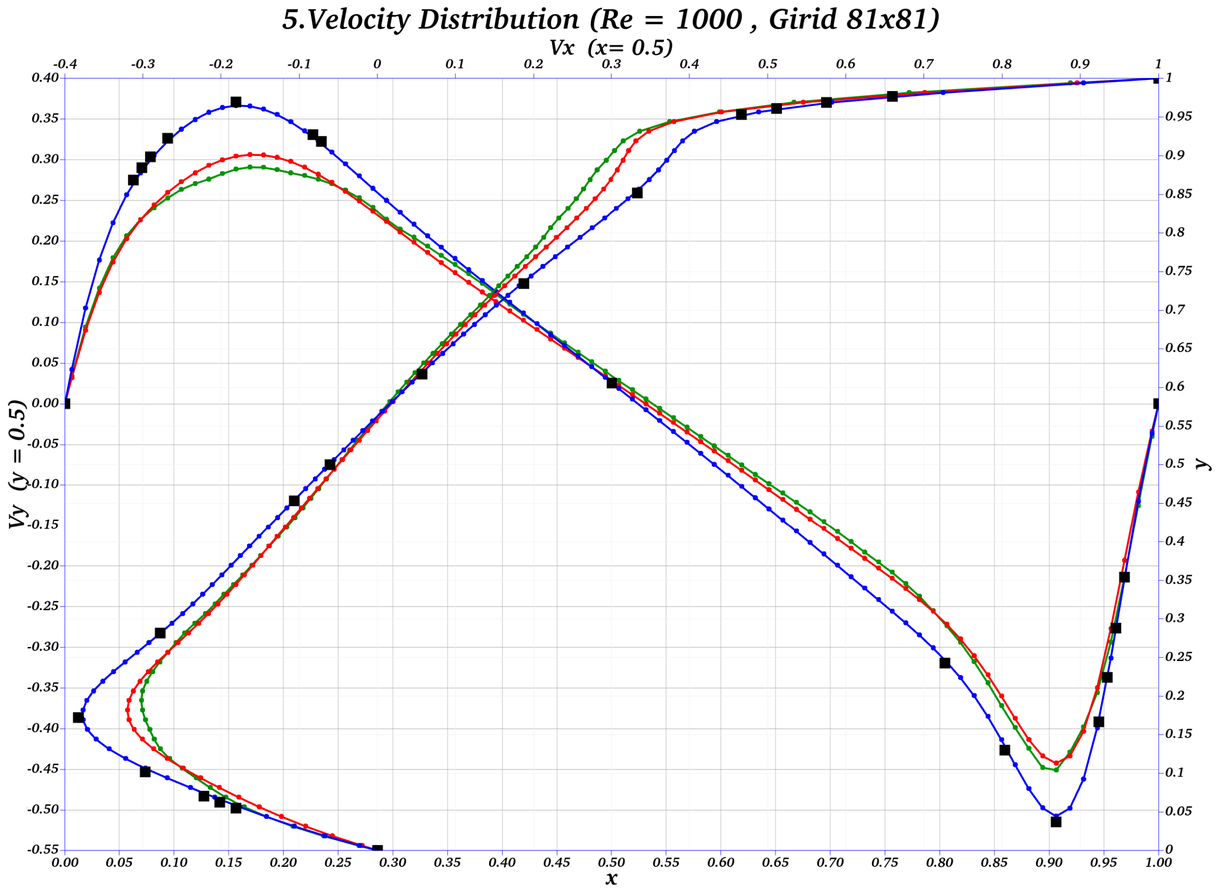
(1) Centerline velocity comparison under the same discretization scheme



5. Result and Discussion

(2) Centerline velocity comparison under the same R_e number

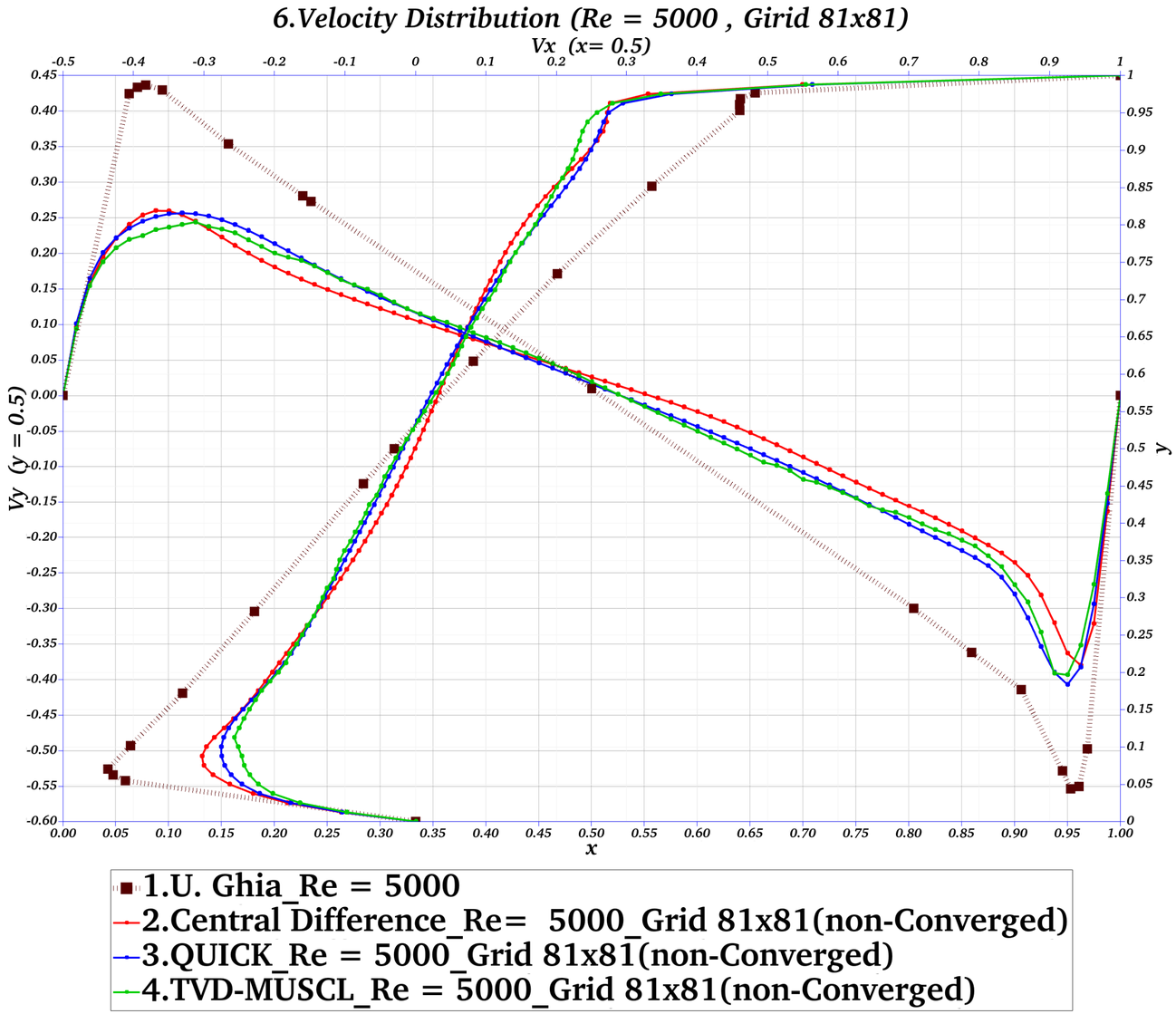




5. Result and Discussion

(2) Centerline velocity comparison under the same R_e number

1. Re = 5000
2. Problem : non-Converged
3. Current mass flow rate : 0.007
4. How long can get the converged solution ? 4 days

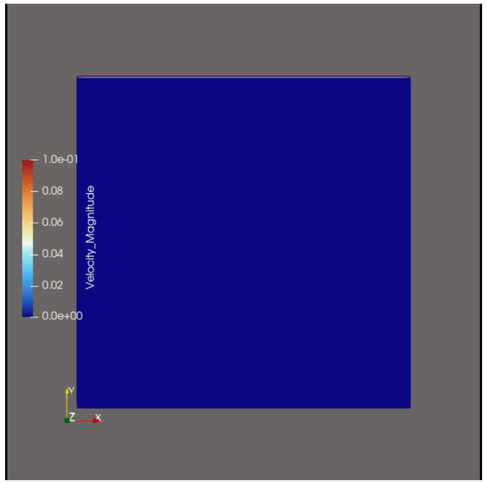


6. Conclusion

(1) For the Central Difference Scheme, I use FDM to solve the unsteady Navier-Stokes equations. The detailed methodology will be presented in the next report.

(2) For QUICK (Re = 100, 1000) and TVD (Re = 100), I use FVM to solve the steady N-S equations, and convergence is successfully achieved.

(3) However, for QUICK (Re = 5000) and TVD (Re = 1000, 5000), I encounter convergence difficulties despite progressively reducing the relaxation factor. The velocity field exhibits an oscillatory, checkerboard-like distribution pattern. Finally I added deferred correction method and successfully modified this condition.



A. Appendix

Step	
1	$$(a_P)_{i,J}^{*\,N} u_{i,J}^{*\,N+\frac{1}{2}} + (a_{WW})_{i,J}^{*\,N} u_{i-2,J}^{*\,N} + (a_W)_{i,J}^{*\,N} u_{i-1,J}^{*\,N} + (a_E)_{i+1,J}^{*\,N} u_{i+1,J}^{*\,N} + (a_{EE})_{i+2,J}^{*\,N} u_{i+2,J}^{*\,N} + (a_{SS})_{i,J}^{*\,N} u_{i,J-2}^{*\,N} + (a_S)_{i,J}^{*\,N} u_{i,J-1}^{*\,N} + (a_N)_{i,J}^{*\,N} u_{i,J+1}^{*\,N} + (a_{NN})_{i,J}^{*\,N} u_{i,J+2}^{*\,N}$$ $$= \frac{-\left(p_{I,J}^{*\,N} - p_{I-1,J}^{*\,N}\right)}{\rho \delta x} \delta x \delta y$$
2	$$\left(\frac{\delta y}{\rho (a_P)_{i+1,J}^{*\,N+\frac{1}{2}}} + \frac{\delta y}{\rho (a_P)_{i,J}^{*\,N+\frac{1}{2}}} + \frac{\delta x}{\rho (b_P)_{I,j+1}^{*\,N+\frac{1}{2}}} + \frac{\delta x}{\rho (b_P)_{I,j}^{*\,N+\frac{1}{2}}}\right) p_{I,J}^{'\,N+1} = \left(\frac{\delta x}{\rho (b_P)_{I,j}^{*\,N+\frac{1}{2}}}\right) p_{I,J-1}^{'\,N} + \left(\frac{\delta x}{\rho (b_P)_{I,j+1}^{*\,N+\frac{1}{2}}}\right) p_{I,J+1}^{'\,N} + \left(\frac{\delta y}{\rho (a_P)_{i,J}^{*\,N+\frac{1}{2}}}\right) p_{I-1,J}^{'\,N} + \left(\frac{\delta y}{\rho (a_P)_{i+1,J}^{*\,N+\frac{1}{2}}}\right) p_{I+1,J}^{'\,N}$$ $$+ u_{i,J}^{*\,N+\frac{1}{2}} - u_{i+1,J}^{*\,N+\frac{1}{2}} + v_{I,j}^{*\,N+\frac{1}{2}} - v_{I,j+1}^{*\,N+\frac{1}{2}} = 0$$
3	$$p_{I,J}^{N+1} = p_{I,J}^{*N} + p_{I,J}^{'\,N+1}$$
4	$$(a_P)_{i,J}^{*\,N+\frac{1}{2}} u_{i,J}^{'\,N+1} = \frac{-\left(p_{I,J}^{'\,N+1} - p_{I-1,J}^{'\,N+1}\right)}{\rho \delta x} \delta x \delta y$$
5	$$u_{i,J}^{N+1} = u_{i,J}^{'\,N+1} + u_{i,J}^{*\,N+\frac{1}{2}}$$
6	$$A_e u_{i+1,J}^{*\,N+\frac{1}{2}} + A_w u_{i,J}^{*\,N+\frac{1}{2}} + A_n v_{I,j+1}^{*\,N+\frac{1}{2}} - A_s v_{I,j}^{*\,N+\frac{1}{2}} \stackrel{?}{=} 0$$
7	$$u_{i,J}^{*\,N+1} = u_{i,J}^{N+1}$$
8	$$p_{I,J}^{*\,N+1} = p_{I,J}^{N+1}$$